

ISRO Lunar Mission

Estimated Budget & Component Pricing Proposal

Mission Executive Summary: Total Estimated Budget

The estimated total budget for a modern ISRO lunar mission featuring a main rover and an auxiliary hopping scout is **₹700 crore to ₹900 crore** (approximately **\$90 million to \$120 million**).

This comprehensive mission budget uses the Chandrayaan-3 mission as a baseline, which cost roughly ₹615 crore (~\$75 million) for the combined orbiter, lander, rover, and launch. The addition of a hopping scout rover contributes an estimated ₹50 to ₹150 crore in hardware, paired with ₹50 to ₹100 crore for integration testing and ₹100 to ₹200 crore for launch operations.

Level 1: Launch, Integration, and Overhead Costs

Before detailing the rover hardware, spacecraft must undergo rigorous qualification and secure a launch vehicle.

- **Launch Vehicle (LVM3 / Artemis-class):** Costs run roughly \$50 million to \$100 million per launch, or roughly ₹100 crore to ₹200 crore.
- **Integration and Testing Overhead:** Bringing space-grade components to flight standards and ensuring redundancy adds a 20% to 30% overhead to the mission. For context, ISRO budgeted a ₹150 crore (~\$16 million) reserve for operations and test equipment during Chandrayaan-3.

Level 2: Main Rover Component Pricing

The primary rover relies on radiation-hardened, space-qualified hardware. Pricing for these core subsystems scales highly depending on power and redundancy requirements.

1. Structure, Mobility, and Actuators

- **Space-Grade Brushless DC Motors:** ~\$5,000 to \$20,000 per motor.
- **Specialized Actuators (e.g., Robotic Drills):** ~\$100,000 per unit, which includes development costs.

2. Power & Thermal Management

- **Space-Grade Solar Panels:** ~\$100 to \$200 per watt. A standard 100 W rover system costs approximately \$10,000 to \$20,000.
- **Radiation-Tolerant Li-Ion Batteries:** ~\$1,000 per Ampere-hour (Ah). A 50 Wh battery pack costs approximately \$50,000.
- **Radioisotope Heater Units (RHU):** ~\$40,000 to \$42,000 each for passive thermal control.
- *(Note: Full Radioisotope Thermoelectric Generators (RTGs) for continuous power are excluded from this budget as they cost tens of millions of dollars each.)*

3. Avionics, Guidance, and Communications

- **Flight Processors and IMUs:** Rad-hardened redundant flight processors and Inertial Measurement Units (IMUs) cost between \$100,000 and \$1,000,000 each.
- **Primary Communications:** UHF/X-band radios and amplifiers range from \$100,000 to \$1,000,000.
- **Secondary Relay Radio:** Adding a supplementary radio to the main rover to relay communications from a hopper scout costs roughly \$50,000.

4. Scientific Payloads

- **Space-Qualified Imagers and Spectrometers:** Instruments like an Alpha Particle X-ray Spectrometer (APXS) or Laser-Induced Breakdown Spectroscopy (LIBS) typically cost over \$1,000,000 per instrument.

Level 3: Auxiliary Hopping Scout Rover Pricing

Designed to augment the main rover by exploring permanently shadowed craters, this 3–8 kg scout requires highly specialized but miniaturized components.

- **Hopping Mechanism:** A gas actuator or spring-based one-leg hop mechanism is estimated at under \$500,000 in development and hardware.
- **Attitude Control:** An IMU and controller chip to manage orientation costs in the tens of thousands of dollars.
- **Sensors and Mapping:** A suite including Flash LiDAR, stereo cameras, and environmental sensors ranges from \$500,000 to \$1,000,000. Commercial Off-The-Shelf (COTS) 3D LiDAR units alone run from \$100,000 to several hundred thousand dollars.
- **Localization Beacons:** A UWB or RF beacon to triangulate the scout's position relative to the main rover costs approximately \$10,000.